BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

MATHEMATICS PART-I — (Re-appear) — (VERSION A) — SECTION-A

Time Allowed: 20 minutes | **Maximum Marks:** 20 | **Paper Code:** BN-XI-XXII-II

Note: Note: There are THREE sections in this papers i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. $z \cdot \bar{z} = ?$

(A) $|z|$ (B) $|z|^2$

(C) $Re(z)$ (D) $Im(z)$

Q2. Area of triangle is:

(A) $|\vec{a} \times \vec{b}|$ (B) $|\vec{a} + \vec{b}|$

(C) $\frac{1}{2}|\vec{a} \times \vec{b}|$ (D) None of these

Q3. The modulus of i = (0,1) is equal to:

(A) -1 (B) 0

(C) 1 (D) ∞

Q4. A square matrix A is said to be singular if:

(A) $A = 0$ (B) $A \neq 0$

(C) $|A| = 0$ (D) $|A| \neq 0$

Q5. If A = [[1,-2,5],[3,0,1],[5,2,0]] then the cofactor of A_{13} is:

(A) 10
(B) 7
(C) 5
(D) 6

Q6. A vector whose magnitude is one is called:

(A) Unit vector
(B) Null vector
(C) Negative vector
(D) None of these

Q7. The angle (θ) between $\vec{a}$ and $\vec{b}$ is 0 or π, then $\vec{a}$ and $\vec{b}$ are:

(A) parallel
(B) Perpendicular
(C) Collinear
(D) Both A and C

Q8. $\vec{i}\cdot(\vec{k}\times\vec{j})$ is equal to:

(A) 0
(B) 1
(C) -1
(D) None of these

Q9. The [illegible in source scan] term of the H.P. 1/3, 1/5, 1/7, ... is:

(A) 1/12
(B) 1/13
(C) 1/14
(D) 1/15

Q10. The series 2+2+2+2+2+... is:

(A) Convergent
(B) Divergent
(C) Finite
(D) Bounded

Q11. $0.2\overline{1}$ in fraction form is:

(A) 7/31
(B) 7/32
(C) 7/33
(D) 7/34

Q12. (6 / 2,3,1) = ?

(A) 36
(B) 1
(C) 60
(D) 12

Q13. The sum of the coefficients in the expansion of $(x + y)^5$ is:

(A) 32
(B) 64
(C) 98
(D) 106

Q14. The period of sin θ is:

(A) π
(B) π^2
(C) 2π
(D) -π

Q15. $1 + 3 + 5 + ... + (2n - 1) = ?$

(A) n^2 (B) n^3

(C) 2^n (D) 3^n

Q16. Which of the function is (1-1)?

(A) $f(x) = |x|$ (B) $f(x) = x^2$

(C) $f(x) = x^3$ (D) $f(x) = \sin x$

Q17. $\sin(\pi - \theta)$ is equal to:

(A) $\sin \theta$ (B) $\cos \theta$

(C) $-\sin \theta$ (D) $-\cos \theta$

Q18. Range of $f(x) = \cos x$ is:

(A) R (B) [-1,1]

(C) (-1,1) (D) $(0, \infty)$

Q19. The period of $\cos 6x$ is:

(A) $\pi/2$ (B) $\pi/3$

(C) $\pi/4$ (D) 2π

Q20. Absolute value of i is:

(A) i (B) 1

(C) -1 (D) ± 1

MATHEMATICS PART-I — (Re-appear) — SECTION-B

Time Allowed: 02:40 Hrs | **Maximum Marks:** 50

Note: Note: Attempt section B & C accordingly.

Q2. (Marks 50) Q.No.2: Attempt any TEN parts of the following. All parts carry equal marks.

i. Divide $2 + 3i$ by $5 - 3i$.

ii. Find x if the determinant $|x\ 2\ 3; 0\ -1\ 1; 0\ 4\ 5| = 9$.

iii. Find the unit vector $\hat{a}$ if $\vec{a} = 3\hat{i}$.

iv. Insert three arithmetic means between 2 and 9.

v. Find five more terms of the H.P. 1/3, 1, -1, ...

vi. In how many ways can a police department arrange eight suspects in a line up?

vii. How many diagonals can be drawn a plan figure of 8 sides?

viii. Prove that $2^{2n} - 1$ is a multiple of 3 for all positive integers.

ix. Show that g(x) = 0 is both even and odd.

x. Solve the inequality $3x - 2 \geq 8 + 5x$.

xi. Prove that $(\sin\theta - \cos\theta)^2 = 1 - \sin 2\theta$.

xii. Find the domain and period of sin 2x.

xiii. Show that $\cos(\pi/2 - \theta) = \sin\theta$

SECTION-C

Maximum Marks: 30

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03.

a) Find the multiplicative inverse of z = 1 + 2i.

b) If A and B are non-singular matrices, then show that $(A^{-1})^{-1} = A$

Q04.

a) Find a unit vector perpendicular to both $\vec{a} = \vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b} = -2\vec{i} + \vec{j} - 3\vec{k}$.

b) Find the value of n, $(a^{n+1} + b^{n+1})/(a^n + b^n)$ is the arithmetic mean between a and b.

Q05.

a) Show by mathematical induction that $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + \ldots + 1/(n(n+1)) = n/(n+1)$.

b) Show that $\sin 5\theta + 2\sin 3\theta + \sin\theta = 4\sin 3\theta\cos^2\theta$.

Q06.

a) $r_1r_2r_3 = rs^2$.

b) Find x, if $\sin^{-1}(1/2) = \pi/2 - x$.

GROUP – II

MATHEMATICS (PART-I) — (Fresh) — (Version-B) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 20 | **Paper Code:** BN-XIXXII-II

Note: Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type

question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. In the expansion of $(a+b)^n$ if n is even, then there will be _____ middle term(s).

(A) 2
(B) n+1
(C) 1
(D) N

Q2. The amplitude of y = sin 6θ is _____

(A) 3
(B) 6
(C) 2π
(D) 1

Q3. 1/19, 1/14, 1/10, 1/6, _____ is.

(A) Geometric sequence
(B) Arithmetic sequence
(C) Asymptotic sequence
(D) Harmonic sequence

Q4. $\Sigma(j=1 \text{ to } n)\ j^2$ = _____

(A) n(n+1)(2n+1)/3
(B) n(n+1)(2n+1)/6
(C) n(n+1)/2
(D) $n^2(n+1)^2/4$

Q5. If 2,5,8,.....is an A.P., then a_{15} = _____

(A) 44
(B) 41
(C) 47
(D) 50

Q6. Geometric mean of -6 and -216 is _____

(A) -11
(B) ±36
(C) -222
(D) -210

Q7. Period of cos 6x _____

(A) π/2
(B) 2π
(C) [illegible in source scan]
(D) [illegible in source scan]

Q8. If C(n,8) = C(n,12) then n = _____

(A) 18
(B) [illegible in source scan]
(C) 17
(D) 9

Q9. If α, β, γ are the direction angles of $\vec{r}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$ = _____

(A) -1
(B) 0
(C) 1
(D) 3

Q10. Maximum value of y=1+2sinθ, is _____

(A) [illegible in source scan]
(B) 1
(C) 2
(D) 3

Q11. $\text{Tan}^{-1}(x/y) - \text{Tan}^{-1}((x-y)/(x+y)) =$ _____

(A) π/2 (B) π/3

(C) π/4 (D) 3π/4

Q12. $(-i)^{23} =$ _____

(A) -1 (B) -i

(C) i (D) 1

Q13. For any complex number z if $-z = \bar{z}$, then z is _____

(A) Purely imaginary (B) Real

(C) 1 + 0i (D) None

Q14. If $\vec{a} = 3i - 5j$, $\vec{b} = -2i + 3j$, then $|\vec{a}/\vec{b}| =$ _____

(A) 34/13 (B) 13/34

(C) √(34/13) (D) -√(34/13)

Q15. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\varphi = 60°$, then $\vec{a}\cdot\vec{b} =$ _____

(A) √(3/2) (B) 7

(C) 1/2 (D) 6

Q16. With usual notation $rr_1r_2r_3 =$ _____

(A) abc/Δ (B) Δ/abc

(C) Δ^2 (D) Δ

Q17. If A = [[1,-2,5],[3,0,-1],[5,2,0]], then $A_{13} =$ _____

(A) 6 (B) 5

(C) -6 (D) 5

Q18. The area of the inscribed circle is given by _____

(A) 2πr (B) $(1/3)\pi r^2$

(C) $(1/2)\pi r^2$ (D) πr^2

Q19. The domain of f(x) = (2-x)/√(x+2) is _____

(A) [-4,2) (B) (-2,2)

(C) (-2,2] (D) All of these

Q20. The period of 3 sin(x/3) is = _____

(A) 6π (B) 3π

(C) π (D) 2π

MATHEMATICS (PART-I) — (Fresh) — SECTION –B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 50

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 50) Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. Show that for any complex number z, $z \cdot \bar{z} = [R(z)]^2 + [I(z)]^2$

ii. Show that $\bar{z} = -z$ if and only if z is pure imaginary.

iii. Show that the determinant $|1\ a\ a^2;\ 1\ b\ b^2;\ 1\ c\ c^2| = (a-b)(b-c)(c-a)$.

iv. Show that $\sqrt{(rr_1r_2r_3)} = \Delta$

v. Find 't' such that the vectors 2i - j + 2k and 3i + 2tj are orthogonal.

vi. If A is square matrix of order 3 then show that $A - A^t$ is skew symmetric.

vii. Sum the series upto n terms $1^2 + 3^2 + 5^2 + 7^2 + ...$

viii. For the vectors $\vec{a} = 3i + 2k$, $\vec{b} = i + 2j + k$, $\vec{c} = -j + 4k$ verify that $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{b} \cdot \vec{c} \times \vec{a} = \vec{c} \cdot \vec{a} \times \vec{b}$

ix. A coin is tossed 6 times in succession. What is the probability that at least one head occurs.

x. Find the middle term in the expansion of $(a/2 + bx)^8$

xi. Sketch the graph of $f(x) = -x^2 + 1$.

xii. Prove that $\sin 2x/\sin x - \cos 2x/\cos x = \sec x$

xiii. The angle of elevation of the top of a post from a point on level ground 38m away is 33.23°. Find the height of the post.

xiv. Find the maximum and minimum value of $y = 3 + 2\cos(3\theta - 2)$

SECTION –C

Maximum Marks: 30

> Note: Note: Attempt any THREE of the following. All questions carry equal marks.

Q3. Find the maximum and minimum values of the function $f(x,y) = 7x + 21y$ subject to the constraints $2x+y \geq 2$, $2x + 3y \leq 6$, $x+2y \leq 8$, $x \geq 0$, $y \geq 0$.

Q4. If $y = 3/4 + 3\cdot5/(4\cdot8) + 3\cdot5\cdot7/(4\cdot8\cdot12) + \ldots$ Show that $y^2 + 2y - 7 = 0$.

Q5.

a) Solve the simultaneous linear equation with complex coefficients $3z+(2+i)w=11-i$ and $(2-i)z-w = -1+i$

b) Find the Echelon form of [[1,0,-2],[2,1,-2],[3,2,3]]

Q6.

a) if $\cos\theta = -3/7$ and terminal ray of θ is in 3rd quadrant, then find $\sin \theta/2$.

b) Find the area of a triangle ABC. $a=15$, $b=80$, $\gamma=38°$.